

Design of Experiments

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Variables

- Numerical
 - Continuous
 - Discrete - known interval
- Categorical
 - Regular categorical
 - Ordinal - interval not implied
 - Dichotomous - binary, Bernoulli
- Role
 - Explanatory - the factor(s) changed
 - Response - response to explanatory
- Misc.
 - Factors - condition imposed
 - Treatment/Level - the magnitude of the factor (concentration, etc)

Terminology

- Skew left - mean < med. Skew right - mean > med
- Kurtosis - deviation from normal due to tail thickness
- Error types

		Decision	
		reject H_0	accept H_0
Truth	$H_0 = \text{true}$	Type 1	correct
	$H_0 = \text{false}$	correct	Type 2

t-test

Standard deviation is variance² If n is big, or Greek letters, or population parameters, use Z-test. Otherwise:

$$\text{Single Group Mean} \quad t = \frac{\bar{x} - (\text{null})}{s/\sqrt{n}} \quad df = n - 1$$

$$\text{Compare Two Means} \quad t = \frac{\bar{x}_1 - \bar{x}_2 - (\text{null})}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

$$df_{\max} = n_1 + n_2 - 2 \quad df_{\min} = \min(n_1 - 1, n_2 - 1)$$

$$\text{Paired/Dependent} \quad t = \frac{\bar{x}_{\text{dif}} - (\text{null})}{s_{\text{dif}}/\sqrt{n}} \quad df = n - 1$$

Confidence Interval

Lower	$CI_L = \bar{x} - t_{\alpha}^* SE$	$(-\infty, \#)$	$SE = s/\sqrt{n}$
Upper	$CI_U = \bar{x} + t_{\alpha}^* SE$	$(\#, \infty)$	$SE = s/\sqrt{n}$
Two-Tailed	$CI_2 = \bar{x} \pm t_{\alpha/2}^* SE$	$(\#, \#)$	$SE = s/\sqrt{n}$

If (null) is within the CI, the sample likely came from the same distribution. (Use t^* from the table, not the t calculated).

Hypothesis Steps

- Set hypotheses
- Choose correct test statistic
- Check assumptions and conditions
- State decisions and draw picture
- Calculate test statistic
- Make decision and interpret in context with words

p-value is the probability that what you saw happened if the null is true. If p is low, the null must go.

Regression

$$\begin{array}{ll} \text{Estimate} & \hat{y} = b_0 + b_1 x_1 + \dots + b_n x_{n-1} \\ \text{Truth} & y = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_{n-1} + \varepsilon \end{array}$$

Source	df	SS	MS	F - stat	F*
Regres.	df_R	SS_R	MS_R	F	table
Error	df_E	SS_E	MS_E		
Total	df_T	SS_T			

$$df_E = df_T - df_R \quad MS_R = \frac{SS_R}{df_R} \quad MS_E = \frac{SS_E}{df_E}$$

$$F = \frac{MS_R}{MS_E} \quad df_T = df_R + df_E \quad p = \# \text{ predictors}$$

$$R = \frac{SS_R}{SS_T} \quad R_{adj}^2 = 1 - \left(\frac{SS_E}{SS_T} \right) \left(\frac{n-1}{n-p-1} \right)$$

ANOVA and Pals

- Completely Random Design (CRD): Entirely randomized, everywhere
- Fixed Effects Models: can only be used to conclude about the exact levels used
- Random Effects Models: can use to infer about entire population
- $F = \text{var}(\text{between groups})/\text{var}(\text{within groups}) = \frac{MSG}{MSE}$
- Conditions:
 - Independent, random, $N < 10\%$ of pop.
 - $\sim$ Normal $\rightarrow$ residuals, Shapiro-Wilks, QQ-Plot
 - Var across groups is roughly equal $\rightarrow$ Bartlett's

Single Factor ANOVA

$$\begin{aligned} H_0 : \mu_i = \dots \mu_a & \quad \text{or } \tau_i = 0 \\ H_A : \mu_i \neq \mu_{i'} & \quad \text{for any pair of } i, i' \end{aligned}$$

$$\begin{aligned} \text{Effects Model:} \quad y_{ij} &= \mu + \tau_i + \varepsilon_{ij} & N &= an \\ \text{Means Model:} \quad y_{ij} &= \mu_i + \varepsilon_{ij} & \mu_i &= \mu + \tau_i \\ \text{Regression Model:} \quad \hat{y} &= \mu + \sum b_i x_i & x_i &= 0 \text{ or } 1 \end{aligned}$$

Factorial ANOVA

$$y_{ijk} = \underbrace{\mu}_{\text{A effects}} + \underbrace{\tau_i}_{\text{B effects}} + \underbrace{\beta_j}_{\text{interaction effects}} + \underbrace{(\tau\beta)_{ij}}_{\text{interaction effects}} + \varepsilon_{ijk} \quad \begin{cases} i = 1, 2, \dots, a \\ j = 1, 2, \dots, b \\ k = 1, 2, \dots, n \end{cases}$$

$$\begin{aligned} A \quad & \begin{cases} H_0 : \tau_1 = \tau_2 = \dots = \tau_a \\ H_A : \text{any } \tau_i \neq 0 \end{cases} \\ B \quad & \begin{cases} H_0 : \beta_1 = \beta_2 = \dots = \beta_b = 0 \\ H_A : \text{any } \beta_i \neq 0 \end{cases} \\ AB \quad & \begin{cases} H_0 : (\tau\beta)_{11} = (\tau\beta)_{12} = \dots = (\tau\beta)_{ij} \\ H_A : \text{any } (\tau\beta)_{ij} \neq 0 \end{cases} \end{aligned}$$

Post-Hoc: multiple t-tests increases type-I error likelihood.
Use correction or Tukey to avoid

Main Effects:

$$A = \frac{1}{2}(Ab + AB - ab - aB) = \text{avg}(\text{factor}_{\text{on}}) - \text{avg}(\text{factor}_{\text{off}})$$

Interaction Effects:

$$AB = \frac{1}{2}(Ab + aB - ab - AB) = \text{avg}(\text{mixed}) - \text{avg}(\text{!mixed})$$

Blocking

Block for uncontrollable factors (eg. height, sex, etc.) or fatigue/learning

Latin Squares:

- Do the sudoku, but also ensure that factor i proceeds factor $i + 1$ the same number of times it follows
- Blocks for two factors. Graeco-Latin Squares blocks for three factors

2^k Design

- Same conditions as ANOVA
- Each factor is high/low or present/not present

$$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$$

	I	A	B	AB
(1)	+	+	+	+
a	+	+	-	-
b	+	-	+	-
ab	+	-	-	+

Sample 2^2 Experimental Design

$$\text{effect size}_j = \frac{1}{\Gamma n}(\text{contrast}_j) \quad \Gamma = 2^{k-1}$$

$$\text{contrast}_j = \sum_{i=\text{rows}} \text{sgn}(i, j) * \sum_i \text{replicates}$$

$$y = \text{grand mean} + \frac{1}{2} \sum (\text{effect size}) * (\text{factor}) \quad \text{factors} = -1, 1$$

Fold-over Designs

After performing a fractional factorial design, the confounding may be undesirable. A fold-over design flips the sign of one (or more) of the generators, and the same fractional factorial design is repeated (assuming 3 factor and higher interactions are non-significant).

$[i]$ = effect size from standard generator

$[i]'$ = effect size after flipping one or more generators

$$\begin{aligned} \text{isolates} &= \frac{1}{2}([i] + [i']) & \text{isolates} &= \frac{1}{2}([i] - [i']) \\ \text{mains} & & \text{2-ways} & \end{aligned}$$

Missing Data

- List-wise Deletion: If any element is missing, omit entire entry
- Pairwise Deletion: Only omit entry if relevant to test
- Hot-deck Imputation: Insert a random value within range of extant data
- Mean Substitution: Insert mean of data
- Regression Substitution: Perform a regression to insert data point(s)
- Maximum Likelihood: Generate and try ≥ 1000 values, do stats to find best entry

DOE Process

